

Subgroup coverage in split conformal classification: marginal validity, Mondrian repair, and small-group failure, with an audit of TabPFN

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Abstract

Split conformal prediction endows any classifier with finite-sample, distribution-free *marginal* coverage, but marginal validity says nothing about how miscoverage is distributed across protected subgroups—the property that matters when prediction sets gate consequential decisions. We study this gap, empirically and through a finite-sample variance argument, on three US Census tasks from folktabs (ACSIIncome, ACSEmployment, ACSPublicCoverage; California), wrapping five predictors in split conformal prediction across three confidence levels, two conformity scores, and four group taxonomies. Marginal coverage conceals systematic subgroup miscoverage: at a 90% target the worst-covered age group receives only 86.2% coverage (gap 0.081), though every model meets the marginal guarantee. Mondrian (group-conditional) calibration repairs this where strata are large—the age gap falls to 0.051 ($p < 10^{-10}$), costing 0.021 in mean set size—but *worsens* it where strata are small: on the race axis, whose smallest calibration stratum holds only ≈ 84 points, the gap rises from 0.046 to 0.057 ($p < 10^{-3}$). We trace this to the finite-sample variance of the in-expectation guarantee: realized per-group coverage fluctuates as $\sqrt{\alpha(1-\alpha)/n_g}$, so the max–min gap grows

as strata shrink. These effects are nearly identical across all five predictors—coverage fairness is governed by the conformal procedure and the group structure, not the model. As the model under audit, TabPFN is the most reliable base learner we tested: better calibrated out of the box than untuned GBDTs (adaptive ECE 0.026 vs. 0.075/0.052), needing no temperature scaling, and yielding the smallest sets. We also flag a silent failure of non-randomized APS on binary tasks, and give support-aware guidance: use LAC, report worst-group coverage, and apply Mondrian only where $n_g \gtrsim 10/\alpha$.

Keywords: conformal prediction, equalized coverage, group-conditional calibration, Mondrian conformal prediction, uncertainty quantification, calibration, algorithmic fairness, tabular foundation models, TabPFN

1. Introduction

Set-valued prediction with coverage guarantees has become a standard tool for trustworthy machine learning. Given any classifier and an exchangeable calibration sample, split (inductive) conformal prediction returns a prediction set that contains the true label with a user-specified probability $1 - \alpha$,
 5 distribution-free and in finite samples [1, 2, 3]. This guarantee, however, is *marginal*: it averages over the whole population. In the high-stakes tabular settings where conformal prediction is increasingly deployed—credit scoring, benefits eligibility, clinical triage—decisions concern individuals who belong
 10 to protected demographic groups, and a guarantee that holds on average can fail conditionally. A 90% marginal guarantee is compatible with, say, 86% coverage for one group and 94% for another, silently concentrating error—less reliable, less informative sets—on the worse-covered group [4]. Whether,

and when, marginal conformal prediction hides such subgroup disparities is
15 the central question of this paper.

The textbook remedy is *Mondrian* (group-conditional) conformal prediction [5, 4], which calibrates a separate quantile within each group and so guarantees coverage *within* every group—“equalized coverage”—rather than only on average. That guarantee, though, holds in expectation over calibration
20 tion draws, and it is purchased with data: each group’s threshold is estimated from only the n_g calibration points falling in that group. When some groups are small—as protected attributes with imbalanced categories routinely are—those thresholds are noisy, and the realized per-group coverage can vary enough that group-conditional calibration *increases* the empirical
25 disparity it was meant to remove. The practically important question is therefore not whether equalized coverage holds in expectation (it does), but when Mondrian stratification helps versus harms in finite samples, and how a practitioner should decide. We give a clean empirical characterization of both regimes together with a simple finite-sample variance argument that
30 predicts which one obtains.

We situate this study in the tabular regime and pair it with a second, timely question: which base learner to trust. For over a decade gradient-boosted decision trees (GBDTs) have been the default for tabular classification [6], but they are now challenged by *tabular foundation models*, most
35 prominently TabPFN, a prior-data fitted network (PFN) that performs supervised learning as a single forward pass of in-context inference and reportedly matches or exceeds tuned GBDT ensembles on small-to-medium tables [7, 8]. The choice of base learner still matters for uncertainty: although

the conformal quantile step repairs *validity* for any model, the calibration
 40 of the underlying probabilities controls set *efficiency*, and good probabilities
 are what make calibrated decisions and tight sets possible. TabPFN’s
 posterior-predictive outputs are often *described* as well calibrated by construction,
 yet we are aware of no matched-protocol audit of (a) how that calibration compares to strong
 45 baselines, (b) how its probabilities behave as conformity scores inside split conformal
 prediction, and (c) whether the coverage it induces is fair across protected groups.
 We therefore adopt TabPFN as the model under audit—a strong, modern base learner
 whose uncertainty we scrutinize alongside the conformal questions above, against XGBoost,
 LightGBM, and an MLP. For GBDTs and conventional neural networks the
 50 calibration literature is mature [9, 10]; for tabular foundation models it is nascent.

We study three binary prediction tasks derived from US Census micro-
 data via the folktables package [11]—ACSIIncome, ACSEmployment, and ACSPublicCoverage
 for California—each carrying protected attributes (sex,
 55 race, age). We compare TabPFN, with and without post-hoc temperature scaling,
 against XGBoost, LightGBM, and an MLP (all with fixed, untuned default hyperparameters)
 under an identical train/calibration/test protocol with five random seeds, and we wrap
 every model in split conformal prediction at three target levels (80%, 90%, 95%) using
 two conformity scores (LAC
 60 and APS), in both marginal and Mondrian (group-conditional) variants.

Our research questions and headline answers are:

- **RQ1 (Calibration).** *Are TabPFN’s native probabilities better calibrated than those of untuned GBDTs and an MLP?* Yes, substantially

so relative to GBDTs: pooled adaptive ECE is 0.026 for TabPFN ver-
65 sus 0.075 (XGBoost) and 0.052 (LightGBM), with paired Wilcoxon
 $p < 10^{-4}$; TabPFN is statistically indistinguishable from the MLP
($p = 0.60$) while being more accurate, and temperature scaling adds
no significant improvement ($p = 0.12$).

- **RQ2 (Conformal validity and efficiency).** *Do all models achieve*
70 *nominal marginal coverage, and which yields the most efficient (small-*
est) prediction sets? With the LAC score, empirical coverage tracks all
three targets to within one percentage point for every model; TabPFN
yields the smallest sets at every level (e.g., 1.296 vs. 1.327–1.363 at
90%; $p < 10^{-3}$ against each baseline). We additionally document that
75 the APS score *without randomization* is degenerate on these binary
tasks, returning the full label set almost surely—a practically impor-
tant negative result.

- **RQ3 (Coverage fairness).** *Does marginal conformal prediction hide*
subgroup miscalibration, and does Mondrian conformal repair it? Marginal
80 coverage conceals an age-related gap of 0.081 at the 90% target (worst-
group coverage 0.862). Mondrian stratification repairs this for the
age axis (gap $\rightarrow$ 0.051, worst-group coverage $\rightarrow$ 0.881, $p < 10^{-10}$)
at a small efficiency cost (+0.021 mean set size), is neutral on the
already-equitable sex axis, but *increases* the empirical gap on the race
85 axis (from 0.046 to 0.057, $p < 10^{-3}$) without improving worst-group
coverage—a finite-sample failure mode driven by small calibration strata.
Crucially, these effects are nearly identical across all five models: cov-

erage fairness is governed by the conformal procedure and the group structure, not by the choice of predictor.

90 Our contributions are: (1) an empirical demonstration, across five predictors and three census tasks, that marginal conformal validity conceals systematic subgroup miscoverage, together with a finite-sample variance characterization of when group-conditional (Mondrian) calibration repairs it—large strata—versus *worsens* it—small strata—including a documented, statistically significant small-group failure on the race axis, and the finding
95 that these effects are essentially model-independent; (2) a practically important negative result on non-randomized APS for binary tasks; (3) the first matched-protocol audit, to our knowledge, of a tabular foundation model’s calibration and conformal efficiency against GBDT and MLP baselines (in
100 their standard untuned configurations), establishing that TabPFN is better calibrated than GBDTs—needing no temperature scaling—and yields the smallest prediction sets; (4) concrete, support-aware deployment guidance (use LAC, report worst-group coverage, and apply Mondrian only where $n_g \gtrsim 10/\alpha$); and (5) an open, fully scripted pipeline with committed per-seed
105 result files and a verification script that recomputes every reported number, enabling exact reproduction.

The remainder of the paper is organized as follows. Section 2 reviews related work and positions our novelty. Section 3 formalizes the calibration metrics, split conformal prediction, the LAC and APS scores, and Mondrian
110 conformal prediction. Section 4 details datasets, models, protocol, and the statistical analysis plan. Section 5 reports results for RQ1–RQ3. Section 6 interprets the findings and derives deployment guidance, Section 7 discusses

threats to validity, and Section 8 concludes.

2. Related work

115 *Conformal prediction..* Split (inductive) conformal prediction converts any scoring model into set-valued predictions with finite-sample, distribution-free marginal coverage under exchangeability [1, 2, 3]. For classification, the LAC/homogeneous score thresholds $1 - \hat{p}_y$ [12], while APS accumulates sorted class probabilities to improve conditional coverage [13]; APS requires
120 tie-breaking randomization to avoid over-coverage, a detail our results show to be decisive on binary tasks.

Group-conditional coverage and fairness in uncertainty quantification.. Equalized coverage asks that prediction sets cover each protected group at the nominal rate, achievable by Mondrian/group-conditional calibration [4, 5].
125 Subsequent work studies class-conditional and group-conditional validity, its finite-sample costs, and approximate conditional coverage [14, 15, 16]. The guarantee is an expectation over calibration draws, and its finite-sample slack and variance grow as strata shrink. A parallel theoretical line shows that *exact* feature-conditional coverage is impossible without strong assumptions,
130 motivating relaxations that target approximate or multivalid conditional coverage at controlled cost [15, 16], and class-conditional variants that share calibration data across many strata to keep per-group counts usable [14]. These methods exist precisely because naive group-conditional calibration is fragile when groups are small—the failure mode we isolate empirically. Yet audits of
135 realized *coverage disparity* have targeted conventional models; none (to our knowledge) includes a tabular foundation model, and the regime in which

plain Mondrian calibration *backfires* on small strata has received little direct empirical characterization. Our RQ3 supplies that characterization together with a closed-form predictor (Section 3.5) for when it occurs—a quantification of when the equalized-coverage guarantee is actually informative, in the
140 approximate-reasoning sense.

Calibration of machine-learning classifiers.. Modern neural networks are frequently miscalibrated and benefit from post-hoc recalibration such as temperature scaling [9]; boosted trees are known to push scores away from the
145 extremes, motivating Platt scaling and isotonic regression [10]. Metric estimation itself is delicate: binned ECE is biased, and adaptive (equal-mass) binning is the recommended estimator [17]. We adopt adaptive ECE as our primary metric and report Brier score and NLL as proper-scoring complements.

Tabular foundation models and prior-data fitted networks.. PFNs meta-train
150 a transformer on synthetic datasets drawn from a prior over data-generating mechanisms, so that supervised prediction on a new table reduces to in-context inference [18]. TabPFN instantiated this idea for small tabular classification [7] and its successor scaled it to larger tables and regression [8].
155 Evaluations of these models have focused overwhelmingly on accuracy and AUC against GBDT baselines [6, 19]; their uncertainty behaviour is usually asserted (via the Bayesian-posterior interpretation) rather than audited.

Benchmarking tabular deep learning under matched protocols.. A recurring lesson of tabular benchmarking is that conclusions are protocol-sensitive—
160 tuning budgets, preprocessing, and split design can reverse rankings [6, 20].

We therefore hold the entire pipeline fixed across models and vary only the predictor.

Novelty. The intersection of these strands is empty: existing TabPFN evaluations do not audit calibration or conformal behaviour against matched base-
 165 lines; the conformal-fairness literature has not examined foundation-model conformity scores; and tabular benchmarks do not report group-conditional coverage. We fill precisely this gap, in the regime where TabPFN is intended to operate (binary/multiclass tasks of moderate size with demographic structure), and we report the failure modes (APS degeneracy, Mondrian small-
 170 group noise) alongside the successes.

3. Background and preliminaries

We consider classification with covariates $X \in \mathcal{X}$, labels $Y \in \mathcal{Y} = \{1, \dots, K\}$ (here $K = 2$), and, for fairness analyses, a protected attribute $A \in \mathcal{A}$. A probabilistic classifier outputs $\hat{p}(y | x)$.

175 3.1. Calibration metrics

A classifier is *top-label calibrated* if $\Pr(Y = \hat{y} | \hat{p}(\hat{y} | X) = c) = c$ for all confidence levels c , where $\hat{y} = \arg \max_y \hat{p}(y | x)$. The expected calibration error partitions test predictions into B bins $\{\mathcal{B}_b\}$ by confidence and computes

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{B}_b|}{n} \left| \text{acc}(\mathcal{B}_b) - \text{conf}(\mathcal{B}_b) \right|, \quad (1)$$

where acc and conf are the mean accuracy and mean confidence within a
 180 bin. We use *adaptive* (equal-mass) binning with $B = 15$ bins as the primary estimator and report equal-width ECE in the released CSVs; we additionally

report the maximum calibration error (MCE), the Brier score $\frac{1}{n} \sum_i \|\hat{p}(\cdot | x_i) - e_{y_i}\|_2^2$, and the negative log-likelihood (NLL), both strictly proper scoring rules. *Temperature scaling* [9] post-processes logits z as $\text{softmax}(z/T)$ with
185 a single scalar $T > 0$ fit on the calibration split by NLL minimization; it is monotone and therefore accuracy-preserving.

3.2. Split conformal prediction

Split conformal prediction [2, 1] uses a held-out calibration set $\{(x_i, y_i)\}_{i=1}^n$, disjoint from training data, and a conformity score $s(x, y)$ measuring how
190 poorly y conforms at x (larger = worse). Let

$$\hat{q} = \text{the } \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\text{-th empirical quantile of } \{s(x_i, y_i)\}_{i=1}^n. \quad (2)$$

The prediction set for a test point x is $C(x) = \{y : s(x, y) \leq \hat{q}\}$. If the calibration and test points are exchangeable, then for any score function and any model,

$$1 - \alpha \leq \Pr(Y_{\text{test}} \in C(X_{\text{test}})) \leq 1 - \alpha + \frac{1}{n+1}, \quad (3)$$

where the upper bound holds when scores are almost surely distinct [21, 3].
195 Coverage is *marginal*: it averages over the population and need not hold conditionally on subgroups. Exchangeability holds by construction in our protocol—calibration and test points are an i.i.d. split of a single dataset—so Eq. (3) is exact here and any subgroup miscoverage we observe is a property of the *conditional* behaviour the marginal bound leaves unconstrained, not a
200 violation of the bound itself. This is what makes the present setting a clean testbed: there is no shift to confound the conditional-coverage question.

3.3. Conformity scores: *LAC* and *APS*

We evaluate two standard classification scores. *LAC* (least ambiguous set-valued classifier) [12] sets

$$s_{\text{LAC}}(x, y) = 1 - \hat{p}(y \mid x), \quad (4)$$

205 which yields the smallest average set size among procedures with marginal validity when probabilities are correct, but offers no conditional-coverage adaptivity. *APS* (adaptive prediction sets) [13] sorts classes by predicted probability, $\hat{p}_{(1)} \geq \dots \geq \hat{p}_{(K)}$, and scores

$$s_{\text{APS}}(x, y) = \sum_{j: \hat{p}_{(j)} > \hat{p}(y|x)} \hat{p}_{(j)} + u \cdot \hat{p}(y \mid x), \quad u \sim \text{Unif}(0, 1), \quad (5)$$

where the randomization term u is essential for exact validity; the deterministic variant ($u \equiv 1$, i.e., including the full mass of the label’s own class) is 210 conservative. Our implementation uses the deterministic variant (the score accumulates sorted class probabilities up to and including the true label’s own mass, with no tie-breaking randomization), and Section 5.2 shows this is not a cosmetic detail on binary tasks.

215 3.4. *Mondrian (group-conditional) conformal prediction and equalized coverage*

Given a partition of the population by a taxonomy $g(x, a) \in \mathcal{G}$ (here: the protected attribute, or the predicted class), Mondrian conformal prediction [5] computes the quantile in Eq. (2) *separately within each group*, using the 220 n_g calibration points of group g . This yields the group-conditional guarantee

$$\Pr(Y \in C(X) \mid g(X, A) = g) \geq 1 - \alpha \quad \text{for every } g \in \mathcal{G}, \quad (6)$$

i.e., *equalized coverage* in expectation [4]. The price is statistical, and it is the crux of our RQ3. Each group’s threshold $\hat{q}_g$ is estimated from only $n_g \ll n$ calibration points; and because the realized per-group coverage on the test set is an average of (approximately) Bernoulli($1 - \alpha$) indicators, its standard
225 deviation about the nominal level is

$$\text{sd}(\hat{c}_g) \approx \sqrt{\frac{\alpha(1-\alpha)}{n_g}}. \quad (7)$$

The empirical max–min coverage gap over $|\mathcal{G}|$ groups is therefore dominated by the smallest stratum, growing like $1/\sqrt{\min_g n_g}$, and the guarantee’s finite-sample slack $1/(n_g + 1)$ likewise grows as groups shrink. Equalized coverage *in expectation* (Eq. (6)) is thus fully compatible with a *larger realized* gap
230 than marginal calibration produces whenever strata are small—the effect our RQ3 isolates and Eq. (7) quantifies.

3.5. When does Mondrian help? A finite-sample prediction

Whether group-conditional calibration reduces or inflates the realized coverage gap can be predicted, to leading order, from two competing quantities,
235 and this prediction is the conceptual core of our paper. Write the per-group test coverage as $\hat{c}_g$. Under *marginal* calibration all groups share the single global threshold $\hat{q}$, so $\hat{c}_g \approx (1 - \alpha) + b_g$, where b_g is a *structural bias*: the systematic over- or under-coverage the global threshold inflicts on group g because that group’s score distribution departs from the pooled one. The
240 marginal coverage gap is therefore $\max_g b_g - \min_g b_g$ up to a small sampling term—it is set by the *spread of structural biases* and is essentially independent of how large the calibration set is.

Under *Mondrian* calibration each group receives its own threshold $\hat{q}_g$. This removes the bias ($\mathbb{E}[\hat{c}_g] = 1 - \alpha$ for every group) but injects estimation
 245 noise. A standard result on the distribution of split-conformal coverage—conditional on the calibration draw, the coverage is Beta-distributed about the nominal level [3]—gives, for a group calibrated on n_g points, a per-group coverage standard deviation

$$\sigma_g \approx \sqrt{\frac{\alpha(1-\alpha)}{n_g}}. \quad (8)$$

Treating the $|\mathcal{G}|$ group coverages as approximately independent fluctuations
 250 about $1 - \alpha$, the expected max – min range scales as $\sigma_{\min} \sqrt{2 \ln |\mathcal{G}|}$ and is dominated by the smallest stratum. Mondrian thus replaces a *bias-driven* gap with a *noise-driven floor*,

$$\text{gap}^{\text{Mond}} \gtrsim \sqrt{\frac{\alpha(1-\alpha)}{\min_g n_g}} \sqrt{2 \ln |\mathcal{G}|}. \quad (9)$$

The qualitative prediction is immediate: *Mondrian helps exactly when the marginal bias spread exceeds this noise floor, and hurts when it does not*—
 255 and because the floor grows as $1/\sqrt{\min_g n_g}$, the failure regime is entered by shrinking the smallest group, not by changing the model.

Equations (8)–(9) are quantitatively borne out (Section 5.3). At the 90% level ($\alpha = 0.1$), the age axis has $\min_g n_g = 140$ and $|\mathcal{G}| = 5$, a noise floor of $\sqrt{0.09/140} \sqrt{2 \ln 5} \approx 0.046$; its marginal bias spread is 0.081, comfortably
 260 above the floor, so Mondrian should and does reduce the gap, landing at 0.051—just above the predicted floor. The race axis has $\min_g n_g = 84$ (the Black stratum) and $|\mathcal{G}| = 4$, a floor of $\sqrt{0.09/84} \sqrt{2 \ln 4} \approx 0.054$; its marginal bias spread is only 0.046, *below* the floor, so Mondrian cannot improve on

marginal calibration and instead lifts the realized gap to $0.057 \approx$ the floor.
 265 The sex axis ($\min_g n_g \approx 900$, two groups) has a negligible floor (≈ 0.012)
 and an already-tiny bias (0.012), so neither method has anything to do and
 the two coincide. The same arithmetic explains the trend across confidence
 levels in Table 6: as the target tightens and α falls, σ_g shrinks, the floor
 drops, and Mondrian’s gap follows it down. We stress that Eqs. (8)–(9) are a
 270 leading-order estimate—the Gaussian-range step is heuristic and the groups
 are not exactly independent—but they capture the mechanism and predict
 the sign of the marginal-to-Mondrian change on every axis we study.

3.6. Fairness metrics for prediction sets

For groups $g \in \mathcal{G}$ with empirical per-group coverage c_g and mean set size
 275 $\bar{k}_g$ on the test set, we report the *coverage gap* $\max_g c_g - \min_g c_g$, the *worst-*
group coverage $\min_g c_g$, and the *set-size disparity* $\max_g \bar{k}_g - \min_g \bar{k}_g$. The first
 two summarize *coverage* equity; the last summarizes *efficiency* equity—how
 unevenly the uncertainty “budget” (set size) is distributed across groups. We
 will see in Section 5.3 that these two notions can move in opposite directions
 280 under Mondrian calibration, so reporting both is essential.

4. Methodology

4.1. Datasets and protected attributes

We use three binary prediction tasks from the folktables suite [11], all
 on California ACS microdata (2018 1-Year person-survey release): **ACSIIn-**
 285 **come** (predict income $> \$50k$), **ACSEmployment** (predict employment
 status), and **ACSPublicCoverage** (predict public health-insurance cover-
 age). These tasks were introduced precisely as fairness-aware replacements

for UCI Adult and carry the protected attributes we audit: *sex* (2 groups, derived from the binary SEX field), *race* (from RAC1P, coarsened to four
290 groups—White, Black, Asian, and a merged *Other* stratum collecting all remaining RAC1P codes), and *age* (AGEP binned into five bands with edges at 25, 40, 55, and 65: <25, 25–39, 40–54, 55–64, 65+). ACSPublicCoverage is defined by folktables only for respondents under 65, so it carries four age bands rather than five. In addition to the three protected axes we evaluate
295 a *predicted-class* taxonomy (Mondrian on $\hat{y}$), a standard label-conditional stratification. Per-group calibration-set sizes are reported in Table 1; they are the key to the race-axis result in Section 5.3.

4.2. Preprocessing

Numeric features are median-imputed and categorical features ordinal-
300 encoded (unseen categories mapped to -1), yielding a single `float32` design matrix; only the MLP additionally standardizes features to zero mean and unit variance inside its own pipeline. Each task is randomly subsampled, with the seed, to $n = 8,000$ rows to remain within TabPFN’s supported small-data regime. All five models receive identical feature matrices within a
305 cell. We note explicitly that the protected attributes are *not* withheld: SEX, RAC1P, and AGEAP belong to each folktables task’s standard feature set and are therefore supplied to every model as ordinary inputs; we use them *additionally* for stratified evaluation and Mondrian calibration. Coverage disparities thus arise even though every model can see the protected attribute,
310 which is the realistic setting for these administrative tasks.

Table 1: Per-group calibration-set sizes n_g (seed 0; $n_{\text{cal}} = 2,000$ per dataset, summing each block to that total). Counts shift by at most a few percent across the five seeds. The race axis contains a far smaller minority stratum (Black, $n_g \approx 84\text{--}131$) than any age band, which drives the Mondrian failure of Section 5.3. ACSPublicCoverage has no 65+ band by task definition.

Axis	Group	ACSEmployment	ACSIIncome	ACSPublicCoverage
Sex	Female	1046	961	1112
	Male	954	1039	888
Race	White	1271	1194	1122
	Asian	312	342	308
	Other	333	378	439
	Black	84	86	131
Age	<25	577	231	651
	25–39	393	664	557
	40–54	406	625	450
	55–64	272	340	342
	65+	352	140	—

4.3. Models

We compare five predictors: **TabPFN** (tabpfn package v8.0.7, default classifier checkpoint `tabpfn-v3-classifier-v3_default.ckpt`; default settings, no per-dataset tuning, in keeping with its intended usage); **TabPFN+T**,
315 the same posterior probabilities post-processed by temperature scaling fit on the calibration split; **XGBoost** (300 trees, max depth 6, learning rate 0.1, row/column subsampling 0.9, histogram tree method) and **LightGBM** (300 trees, 31 leaves, learning rate 0.05, row/column subsampling 0.9); and

an **MLP** (two hidden layers of 128 and 64 units, standardized inputs, early
 320 stopping on a 10% internal validation split, at most 300 epochs). All baseline
 hyperparameters are fixed at these common default values; no per-dataset
 hyperparameter search was performed for any model, so the comparison is
 between out-of-the-box configurations rather than tuned ones. Because tem-
 perature scaling is a monotone transform of the probabilities, TabPFN and
 325 TabPFN+T induce (near-)identical conformal prediction sets; their confor-
 mal rows differ only through quantile tie effects (maximum observed differ-
 ence in any coverage-gap cell: 0.0027). We therefore discuss them jointly in
 RQ2–RQ3 and separately only for calibration (RQ1).

4.4. *Train / calibration / test protocol*

330 For each (dataset, seed) cell we draw a label-stratified three-way split
 into proper training (45%), calibration (25%), and test (30%) sets—3,600,
 2,000, and 2,400 rows respectively of the 8,000 sampled per cell (the commit-
 ted results record $n_{\text{test}} = 2,400$, confirming these proportions). The model
 is fit on the training split only; temperature and all conformal quantiles are
 335 computed on the calibration split only; every reported metric is computed
 on the untouched test split. Splits are redrawn with seeds $\{0, \dots, 4\}$, and
 all five models share the identical splits within a cell, enabling paired sta-
 tistical tests. The 25% calibration fraction is deliberate: it leaves enough
 data to fit competitive models while giving the global conformal quantile—
 340 and, crucially, each *Mondrian* sub-quantile—a realistic sample, so that the
 small-group effects we report reflect plausible deployment sizes rather than
 a pathologically starved calibration set. Even at this fraction the smallest
 race stratum receives only ≈ 84 calibration points (Table 1), which is exactly

the regime our analysis targets. Sharing splits across models is what lets us
 345 attribute differences to the predictor (RQ1–RQ2) or to the conformal procedure (RQ3) rather than to sampling, and the five seeds turn each comparison into a paired test over 15 (dataset $\times$ seed) cells.

4.5. Experiment matrix

The full matrix is: 3 datasets $\times$ 5 models $\times$ 5 seeds $\times$ 3 miscoverage levels
 350 $\alpha \in \{0.05, 0.10, 0.20\}$ (targets 95%/90%/80%) $\times$ 2 conformity scores (LAC, APS) $\times$ 4 group taxonomies (sex, race, age, predicted class) $\times$ 2 conformal methods (marginal, Mondrian), yielding 3,600 conformal evaluation rows and 75 calibration rows, all committed to the repository.

4.6. Metrics

355 *Calibration (RQ1)*: accuracy, adaptive ECE (primary), equal-width ECE, MCE, Brier, NLL. *Validity/efficiency (RQ2)*: empirical marginal coverage and mean prediction-set size. *Fairness (RQ3)*: coverage gap, worst-group coverage, and set-size disparity as defined in Section 3, for marginal vs. Mondrian calibration on each protected axis.

360 4.7. Statistical protocol

Unless stated otherwise, point estimates are means over the relevant cells and dispersions are standard deviations across the 15 (dataset $\times$ seed) cells. Pairwise model comparisons use two-sided Wilcoxon signed-rank tests paired at the (dataset $\times$ seed) level ($n = 15$; the smallest attainable two-sided p is
 365 6.1×10^{-5} , which we report as $p < 10^{-4}$). Marginal-vs.-Mondrian comparisons are paired at the (dataset $\times$ model $\times$ seed) level ($n = 75$). Omnibus

Table 2: RQ1 — Accuracy and calibration, mean $\pm$ std over 15 (dataset $\times$ seed) cells.

Lower is better for all metrics except accuracy. Best in **bold**.

Model	Accuracy $\uparrow$	ECE _{ada} $\downarrow$	MCE $\downarrow$	Brier $\downarrow$	NLL $\downarrow$
XGBoost	0.763 ± 0.052	0.075 ± 0.019	0.135 ± 0.022	0.333 ± 0.064	0.516 ± 0.084
LightGBM	0.769 ± 0.049	0.052 ± 0.012	0.100 ± 0.018	0.317 ± 0.057	0.485 ± 0.074
MLP	0.759 ± 0.046	0.025 ± 0.007	0.057 ± 0.022	0.324 ± 0.056	0.486 ± 0.074
TabPFN	0.782 ± 0.046	0.026 ± 0.006	0.063 ± 0.021	0.298 ± 0.056	0.452 ± 0.075
TabPFN+T	0.782 ± 0.046	0.025 ± 0.006	0.054 ± 0.018	0.297 ± 0.056	0.451 ± 0.074

differences across the five models use the Friedman test over the 15 cells with average ranks reported; given only five Tier-A model columns we report ranks descriptively rather than a full Nemenyi critical-difference analysis, which
 370 would require the larger Tier-B suite that is not part of the committed run (see Section 7). We treat $p < 0.01$ as significant in view of the multiplicity of comparisons.

5. Results

5.1. RQ1: TabPFN is natively well calibrated; GBDTs are not

375 Table 2 reports pooled calibration metrics; Table 3 breaks adaptive ECE out by dataset; Fig. 1 shows a representative reliability diagram.

Three observations follow. *First*, both GBDTs are substantially miscalibrated out of the box, with the familiar pattern of over-confidence at high confidence levels (Fig. 1): XGBoost’s pooled adaptive ECE is 0.075 and
 380 LightGBM’s 0.052, against 0.026 for TabPFN. The paired differences are significant at the resolution limit of the test (TabPFN vs. XGBoost and vs.

Table 3: RQ1 — Adaptive ECE by dataset (mean $\pm$ std over 5 seeds).

Model	ACSEmployment	ACSIIncome	ACSPublicCoverage
XGBoost	0.058 ± 0.005	0.069 ± 0.006	0.098 ± 0.009
LightGBM	0.043 ± 0.006	0.049 ± 0.007	0.064 ± 0.010
MLP	0.020 ± 0.006	0.024 ± 0.003	0.033 ± 0.007
TabPFN	0.022 ± 0.005	0.025 ± 0.005	0.030 ± 0.007
TabPFN+T	0.021 ± 0.004	0.025 ± 0.004	0.028 ± 0.007

LightGBM: Wilcoxon $p < 10^{-4}$ in both cases, with TabPFN better in all 15 of 15 cells). *Second*, TabPFN’s native calibration matches a well-regularized MLP (median paired ECE difference -0.001 , $p = 0.60$) while being more
385 accurate (paired accuracy difference $+0.023$, $p < 10^{-3}$) and strictly better on both proper scoring rules (Brier and NLL, $p < 10^{-4}$ against all three baselines). The Friedman omnibus test on adaptive ECE is highly significant ($\chi^2 = 48.5$, $p = 7.5 \times 10^{-10}$) with average ranks TabPFN+T 1.8, MLP 2.0, TabPFN 2.2, LightGBM 4.0, XGBoost 5.0. *Third*, post-hoc temperature scal-
390 ing improves TabPFN only marginally (pooled ECE $0.026 \rightarrow 0.025$; paired $p = 0.12$): within the datasets studied, TabPFN does not need recalibration, which is consistent with—and, to our knowledge, the first matched-baseline statistical evidence for—the posterior-predictive interpretation of PFN outputs.

395 The ranking is consistent across tasks, not an artifact of pooling. TabPFN is the most accurate model on all three datasets individually (accuracy 0.812, 0.814, 0.720 on ACSEmployment, ACSIIncome, ACSPublicCoverage) and the best- or tied-best-calibrated on each (Table 3). ACSPublicCoverage

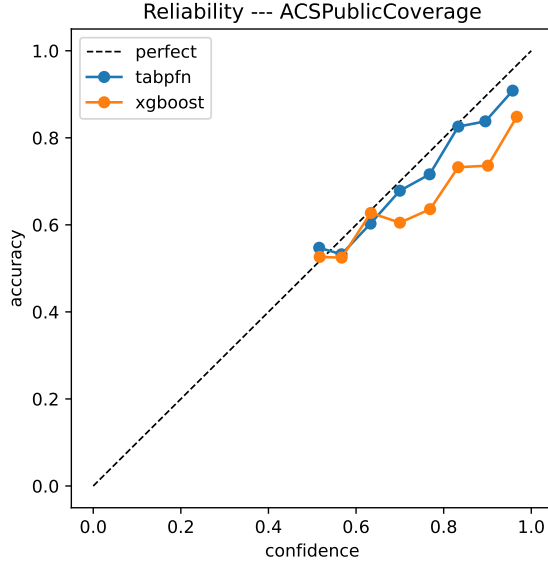


Figure 1: Reliability diagram on ACSPublicCoverage-CA (seed 0). XGBoost (orange) sits below the diagonal at high confidence—systematic over-confidence—while TabPFN (blue) tracks the diagonal closely. Curves use 15 equal-width confidence bins; the dashed line is perfect calibration.

is the hardest task for every model—accuracies cluster near 0.70 and XG-
400 Boost’s adaptive ECE climbs to 0.098—yet TabPFN holds its calibration
there (0.030), so the calibration gap to the GBDTs *widens* precisely where
the problem is hardest and trustworthy uncertainty matters most. The MLP
is the only baseline whose calibration rivals TabPFN’s, but it pays for it in
accuracy: it is the least accurate predictor on two of the three tasks, so its
405 competitive ECE partly reflects less confident predictions rather than better-
ordered ones, whereas TabPFN attains low ECE *and* the highest accuracy
simultaneously.

Table 4: RQ2 — Marginal coverage and mean set size under the LAC score (mean over 15 cells per row). Coverage within one point of target for all models; smaller sets are better. Best set size in **bold**. TabPFN+T is omitted: temperature scaling is monotone, so its conformal sets coincide with TabPFN’s.

Model	Target 80%		Target 90%		Target 95%	
	Cov.	Size	Cov.	Size	Cov.	Size
XGBoost	0.800	1.083	0.899	1.339	0.952	1.534
LightGBM	0.802	1.072	0.901	1.327	0.953	1.519
MLP	0.803	1.095	0.903	1.363	0.951	1.543
TabPFN	0.809	1.058	0.900	1.296	0.950	1.480

5.2. RQ2: All models are conformally valid; TabPFN is the most efficient.

Non-randomized APS is degenerate on binary tasks

410 Table 4 reports marginal coverage and mean set size under the LAC score. Empirical coverage tracks the nominal target for every model and level, as Eq. (3) guarantees: pooled over models, coverage is 0.805 ± 0.013 , 0.901 ± 0.009 , and 0.951 ± 0.007 at the 80%, 90%, and 95% targets respectively. Conformal validity is thus model-agnostic, as theory predicts—miscalibrated
415 GBDT probabilities are repaired by the quantile step.

Efficiency, by contrast, is model-dependent and rewards probability quality. TabPFN produces the smallest prediction sets at every target level; at the 90% level its mean set size of 1.296 improves on LightGBM (1.327), XGBoost (1.339), and the MLP (1.363), with paired Wilcoxon $p = 4.3 \times 10^{-4}$,
420 3.1×10^{-4} , and 6.5×10^{-4} respectively. On a binary task, set size minus one is the singleton-failure rate, so this difference is operationally meaningful:

at the 90% level TabPFN returns an uninformative two-label set on 29.6% of cases versus 36.3% for the MLP—a 6.7-point reduction in abstentions at identical coverage.

425 The efficiency ordering is stable across confidence levels and *widens* as coverage tightens (Table 4): TabPFN’s set-size advantage over the MLP grows from 0.037 at the 80% target to 0.063 at 95%, because sharper, better-calibrated tail probabilities are worth progressively more as the quantile is pushed further into the tail. The ordering also tracks RQ1 calibration quality 430 almost exactly—LightGBM, the better-calibrated GBDT, yields smaller sets than XGBoost at every level—confirming that, once validity is handled by the conformal step, set size is the channel through which probability quality is rewarded. This is the operational case for caring about calibration even behind a distribution-free wrapper that guarantees coverage regardless of the 435 model: validity is free, but efficiency is earned.

A cautionary negative result on APS. Under our deterministic APS implementation, every model at every level returns essentially the full label set: pooled coverage is 1.000 and mean set size is 1.99–2.00 at all three targets. The mechanism is elementary but easy to overlook: on a binary task the 440 deterministic APS score of the true label is the cumulative probability mass *including* the label’s own class, which equals 1 whenever the true label is the lower-ranked class; the calibration quantile at any usable α therefore saturates at (or near) 1, and the resulting sets contain both labels. Randomized APS [13] avoids this by interpolating the final class’s mass. Practitioners 445 wrapping binary tabular classifiers should either use LAC or ensure their APS implementation randomizes; we flag this because deterministic APS is

the default in some libraries and the failure is silent—coverage looks excellent while the sets are vacuous. All fairness analyses below therefore use the LAC score; the corresponding APS rows are retained in the released CSVs but are
 450 uninformative by construction.

5.3. RQ3: Marginal coverage hides subgroup gaps; Mondrian repairs large groups but harms small ones

Table 5 is the core fairness result: per-axis coverage gap, worst-group coverage, and mean set size at the 90% target under LAC, pooled over the
 455 five models and 15 cells, for marginal versus Mondrian calibration. Table 6 shows the coverage gap across all three target levels. Figs. 2–4 display the per-model breakdown.

Marginal validity is not subgroup validity. Although every model satisfies the marginal 90% guarantee, per-group coverage is far from uniform. On the
 460 age axis the pooled coverage gap is 0.081 and worst-group coverage drops to 0.862—i.e., the least-covered age group experiences roughly 38% more mis-coverage than nominal. At the 80% target the age gap widens to 0.151 with worst-group coverage 0.732. The race axis shows a smaller but persistent gap (0.046 at 90%, worst-group coverage 0.875), while the sex axis is essentially
 465 equitable already (0.012). These disparities are a property of the population/score interaction, not of the predictor: per-model gaps under marginal calibration at 90% on age lie in the narrow band 0.078–0.083 across all five models, TabPFN included. A foundation model’s better probabilities buy smaller sets (RQ2) but do not, by themselves, buy equitable coverage.

Table 5: RQ3 — Subgroup coverage fairness at the 90% target (LAC; pooled over 5 models $\times$ 3 datasets $\times$ 5 seeds; marginal vs. Mondrian paired tests with $n = 75$). Gap = max–min per-group coverage; WGC = worst-group coverage. Arrows mark the better direction; p -values are two-sided Wilcoxon for the marginal-vs.-Mondrian contrast.

Axis	Method	Gap $\downarrow$	WGC $\uparrow$	Mean size $\downarrow$	
Sex	Marginal	0.012	0.895	1.324	gap $p = 0.50$ WGC $p = 0.79$
	Mondrian	0.013	0.895	1.327	
Age	Marginal	0.081	0.862	1.324	gap $p = 4.2 \times 10^{-11}$ WGC $p = 2.9 \times 10^{-10}$
	Mondrian	0.051	0.881	1.345	
Race	Marginal	0.046	0.875	1.324	gap $p = 9.3 \times 10^{-4}$ WGC $p = 0.55$
	Mondrian	0.057	0.873	1.334	
Pred. class	Marginal	0.045	0.881	1.324	
	Mondrian	0.013	0.894	1.332	

470 *Mondrian repairs the age axis at a small price..* Group-conditional calibration on age reduces the pooled gap from 0.081 to 0.051 and raises worst-group coverage from 0.862 to 0.881 (both $p < 10^{-9}$, $n = 75$), at a mean set-size cost of +0.021 ($1.324 \rightarrow 1.345$, $p < 10^{-12}$)—about two additional both-label sets per hundred predictions. The same pattern holds at the 80% target (gap
475 $0.151 \rightarrow 0.106$; worst-group coverage $0.732 \rightarrow 0.774$). The predicted-class taxonomy behaves similarly (gap $0.045 \rightarrow 0.013$). For TabPFN specifically, Mondrian-on-age lifts worst-group coverage from 0.862 to 0.881 while its sets remain smaller than any baseline’s marginal sets—the efficiency advantage survives the fairness repair.

Table 6: RQ3 — Coverage gap by target level (LAC; pooled over models, datasets, seeds). Mondrian helps consistently on age, is neutral-to-mixed on sex, and does not help—and at 90% significantly hurts—on race.

	Sex		Age		Race	
Target	Marg.	Mond.	Marg.	Mond.	Marg.	Mond.
80%	0.029	0.021	0.151	0.106	0.056	0.055
90%	0.012	0.013	0.081	0.051	0.046	0.057
95%	0.012	0.010	0.045	0.037	0.034	0.038

480 *Mondrian fails on the small-support race axis..* On race, Mondrian *increases* the empirical gap at the 90% target from 0.046 to 0.057 ($p = 9.3 \times 10^{-4}$) and leaves worst-group coverage unchanged (0.875 vs. 0.873, $p = 0.55$); the same direction holds, non-significantly, at 95%. The effect is consistent across datasets (Employment 0.034 $\rightarrow$ 0.049; Income 0.047 $\rightarrow$ 0.060; PublicCov-
485 erage 0.056 $\rightarrow$ 0.061) and across all five models (Fig. 4). The mechanism is the finite-sample cost identified in Section 3.5: the race taxonomy contains groups with small calibration support—the Black stratum holds only $n_g \approx 84$ –131 calibration points across the three tasks, against 1,100–1,300 for White (Table 1)—so each small group’s threshold $\hat{q}_g$ is a high quantile esti-
490 mated from few points. The group-conditional guarantee of Eq. (6) still holds *in expectation over calibration draws*, but the realized per-group coverage fluctuates with standard deviation $\approx \sqrt{\alpha(1-\alpha)/n_g}$, and the max–min gap statistic grows with the number of groups and with $1/\sqrt{\min_g n_g}$. Equalized coverage in expectation is therefore compatible with—and here empirically
495 produces—*larger* realized disparities than marginal calibration when strata

are small. Sex (two large groups) shows neither benefit nor harm because marginal coverage is already nearly equalized.

The pattern holds across confidence levels. Table 6 traces the coverage gap at all three targets, and the qualitative picture is invariant: age carries the largest marginal gap at every level and is repaired by Mondrian at every level (0.151 $\rightarrow$ 0.106 at 80%, 0.081 $\rightarrow$ 0.051 at 90%, 0.045 $\rightarrow$ 0.037 at 95%), whereas race is never helped—Mondrian matches or slightly worsens it at all three. The magnitudes shrink as the target tightens, exactly as Eq. (8) requires: the per-group standard deviation $\sqrt{\alpha(1-\alpha)/n_g}$ falls with α , so both the structural biases and the Mondrian noise floor contract together. Worst-group coverage tells the same story from the decision-maker’s side: on age it rises under Mondrian at every level (0.732 $\rightarrow$ 0.774, 0.862 $\rightarrow$ 0.881, 0.927 $\rightarrow$ 0.935), confirming that the gap reduction reflects genuinely better coverage for the worst-off group rather than a cosmetic compression of the range.

5.4. The efficiency cost of equalized coverage: a set-size tax

Coverage equity is only half of fairness for prediction sets; the other half is whether the *size* of those sets—the amount of irreducible uncertainty each group is told it faces—is distributed evenly. Table 7 reports the set-size disparity $\max_g \bar{k}_g - \min_g \bar{k}_g$ at the 90% target under LAC, and it exposes a trade-off the coverage-gap numbers alone conceal: *Mondrian calibration improves coverage equity at the expense of efficiency equity*. On the age axis, where Mondrian halves the coverage gap, it simultaneously raises the set-size disparity from 0.236 to 0.369—a 56% increase. The same direction holds on every axis (sex 0.058 $\rightarrow$ 0.086; race 0.123 $\rightarrow$ 0.154; predicted class

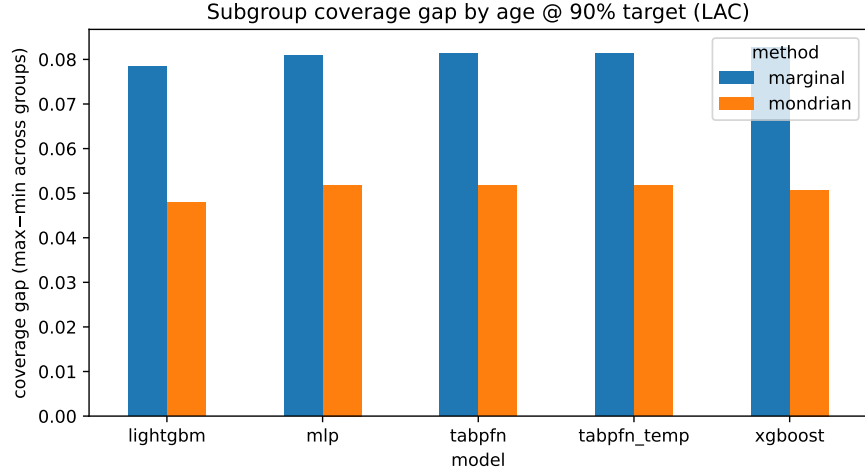


Figure 2: Coverage gap by age at the 90% target, marginal (blue) vs. Mondrian (orange), per model (LAC score). Mondrian reduces the gap by roughly 0.03 uniformly across all five models, TabPFN included.

0.115 $\rightarrow$ 0.157): equalizing coverage forces the harder-to-cover groups to carry visibly larger sets. The effect is mechanical—restoring a group’s coverage to nominal when its scores are less separable requires admitting more labels—but it matters operationally, because a uniform coverage guarantee delivered through systematically larger sets for one group is a different, and arguably more conspicuous, disparity than quiet under-coverage. A practitioner who optimizes only the coverage gap will not see it; one who reports both metrics will.

The finite-sample prediction, validated.. The crossover between the age and race axes is exactly what Section 3.5 forecasts. The Mondrian noise floor of Eq. (9) evaluates to ≈ 0.046 on age (smallest stratum $n_g = 140$, five groups) and ≈ 0.054 on race (smallest stratum $n_g = 84$, four groups). Age’s marginal

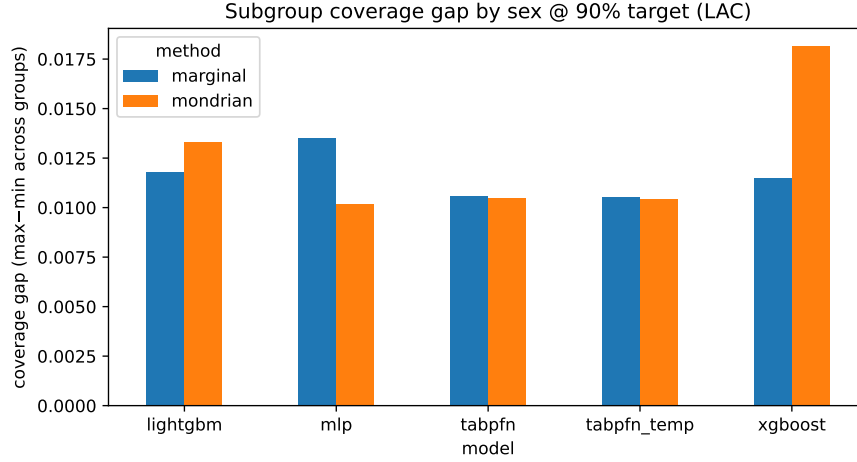


Figure 3: Coverage gap by sex at the 90% target (LAC score). Gaps are an order of magnitude smaller than on age and statistically indistinguishable between methods.

Table 7: RQ3 — Set-size disparity $\max_g \bar{k}_g - \min_g \bar{k}_g$ at the 90% target (LAC; pooled over 5 models $\times$ 15 cells). Mondrian calibration lowers the coverage gap (Table 5) but *raises* the set-size disparity on every axis: coverage equity is bought with efficiency inequity.

Method	Sex	Age	Race	Pred. class
Marginal	0.058	0.236	0.123	0.115
Mondrian	0.086	0.369	0.154	0.157

bias spread (0.081) sits well above its floor, so group-conditional calibration has room to help and the gap drops to 0.051; race’s marginal bias spread
535 (0.046) sits *below* its floor, so calibration noise dominates the bias it removes and the gap rises to $0.057 \approx$ the floor. Sex, with a floor near 0.012 and a bias already at 0.012, sits at the boundary and moves negligibly. That a one-parameter, model-free estimate built only from $\{\alpha, n_g, |\mathcal{G}|\}$ predicts the *sign* of the marginal-to-Mondrian change on all three protected axes—and

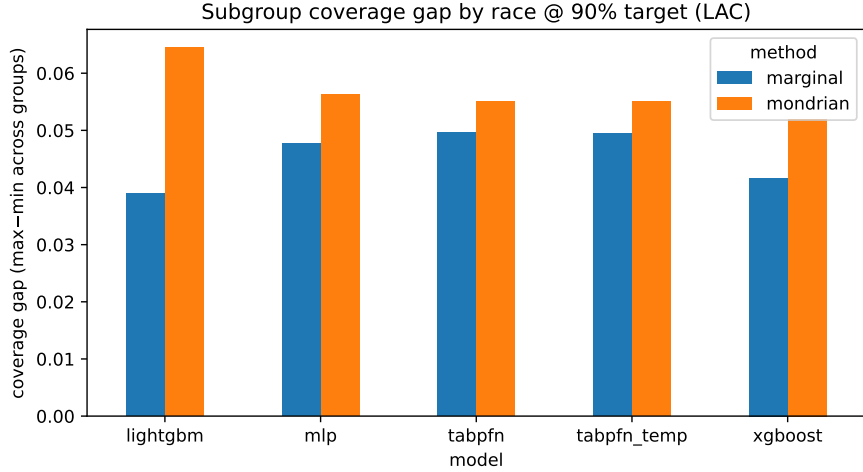


Figure 4: Coverage gap by race at the 90% target (LAC score). Mondrian (orange) *exceeds* marginal (blue) for every model: with small calibration strata, group-conditional thresholds are noisy and realized disparities grow.

540 the approximate magnitude of the Mondrian gap—is, to us, the strongest evidence that the race-axis failure is a generic finite-sample phenomenon rather than an artifact of these particular datasets or predictors.

6. Discussion

What the audit establishes about TabPFN.. Within the studied regime—
 545 binary census tasks, moderate sample sizes, five seeds—TabPFN’s uncertainty is trustworthy on the first two criteria of Section 1: its native probabilities are as calibrated as anything we tested without any post-processing, and they translate into the most efficient conformal sets. Both findings are consistent with the PFN design: a model meta-trained to approximate a
 550 posterior predictive should produce probabilities whose *ordering and sharpness* reflect genuine predictive uncertainty, and LAC efficiency is exactly a

function of probability quality. The practical upshot is that the usual GBDT deployment recipe—train, then Platt/isotonic/temperature recalibrate—can be dropped for TabPFN in this regime; temperature scaling moved pooled
555 ECE by one point in the third decimal and was statistically indistinguishable from the identity.

What it establishes about conformal fairness.. The third criterion—equitable uncertainty—is not a model property at all in our data. Coverage gaps under marginal calibration are nearly identical across the five predictors, and so is
560 the effect of Mondrian stratification. The decision that matters is procedural: *which taxonomy to condition on, given its calibration support.* Our results give an unusually clean empirical demonstration of both sides: conditioning on age (few, large groups, with a real coverage disparity to fix) halves the gap at the cost of 0.02 extra labels per set; conditioning on race (many groups,
565 some very small) backfires, inflating realized disparity without improving the worst group. We emphasize that this is not a contradiction of the equalized-coverage guarantee, which is an expectation over calibration randomness; it is the finite-sample variance of that guarantee made visible. Reporting only the expectation-level guarantee, as is common, would have concealed it.

570 *A diagnostic, not just a diagnosis..* The finite-sample analysis of Section 3.5 turns the race-axis failure from an anomaly into a *predictable* regime, and that carries a concrete payoff: the noise floor of Eq. (9) is computable *before* any group-conditional calibration is run, from quantities a practitioner already has—the target level α , the smallest per-group calibration count
575 $\min_g n_g$, and the number of groups $|\mathcal{G}|$. Comparing that floor to the observed *marginal* coverage gap yields an a priori test of whether Mondrian

stratification can help on a given axis: when the marginal gap is below $\sqrt{\alpha(1-\alpha)/\min_g n_g} \sqrt{2 \ln |\mathcal{G}|}$, group-conditional calibration is expected to inject more noise than the bias it removes, and should be withheld or replaced
 580 by a method with explicit small-sample control [15, 16]. This reframes “equalized coverage” from an unconditional good into a decision with a checkable precondition—a stance closer in spirit to the approximate-reasoning tradition of asking *when* a guarantee is informative than to a blanket recommendation to always condition.

585 *Two axes of fairness that can conflict..* The set-size tax of Section 5.4 sharpens the picture. Coverage equity and efficiency equity are distinct properties, and our data show them moving in *opposite* directions under Mondrian calibration: every axis that gains coverage equity loses efficiency equity (Table 7). There is no single “fair” conformal predictor here, only a trade-off
 590 to be made explicit. For a benefits-eligibility system, an omitted true label (under-coverage) and an inflated, uninformative set (a larger “maybe”) are different harms with different costs, and which to prioritize is a domain question rather than a statistical one. What this paper supplies is the means to see both at once, together with the finite-sample rule that says when paying
 595 the set-size tax actually purchases coverage equity and when it merely redistributes the uncertainty without reducing it.

Deployment guidance.. For practitioners deploying small-to-medium tabular classifiers with demographic structure, our results support the following sequence. (1) If TabPFN’s scale limits permit, prefer it: best accuracy, best
 600 proper scores, no recalibration step. (2) Wrap whatever model is chosen

in split conformal prediction with the *LAC* score; verify that any APS implementation randomizes before using it on low-cardinality label spaces. (3) Audit per-group coverage on a held-out set before choosing between marginal and Mondrian calibration: if a gap exists *and* every group has adequate calibration support, use Mondrian on that axis. As a rule of thumb, a group
605 needs enough points for its $1 - \alpha$ quantile to be stable—on the order of $n_g \gtrsim 10/\alpha$ (about 100 at $\alpha = 0.1$). Our axes straddle this line precisely: the smallest age band (65+, $n_g = 140$ in ACSIncome; Table 1) clears it and Mondrian helps, whereas the smallest race stratum (Black, $n_g = 84$ in
610 ACSEmployment) falls below it and Mondrian backfires. If some groups are small, prefer marginal calibration, merge small strata, or adopt methods designed for approximate conditional validity [15, 16]. (4) Report worst-group coverage alongside marginal coverage; the latter alone certified all our models while one age group received 73% coverage at an 80% target.

615 *TabPFN versus GBDTs, honestly..* Our comparison favours TabPFN on every uncertainty metric, but three caveats temper a blanket recommendation. The tasks sit squarely inside TabPFN’s comfort zone (3,600 training rows per cell, numeric/categorical census features, binary labels); the GBDTs received *no* hyperparameter tuning—they ran with the fixed default settings of
620 Section 4, and a tuning budget (e.g., NLL-based early stopping or a learning-rate/depth search) would likely close part of the calibration gap; and GBDT inference is orders of magnitude cheaper at scale. Where these constraints bind, the correct reading of our results is narrower but still useful: GBDT probabilities should not be consumed raw, and conformal wrapping equalizes
625 validity (though not efficiency) across model families.

7. Limitations and threats to validity

External validity. All results derive from three binary tasks on Californian ACS microdata; conclusions are scoped to this regime and should not be extrapolated to multiclass problems, other states or survey years, non-census domains, or tables beyond TabPFN’s supported size. The planned Tier-B
630 evaluation over a broader OpenML suite, including Friedman–Nemenyi ranking across heterogeneous datasets, was not part of the committed results and remains future work. *Construct validity.* ECE estimates depend on binning; we mitigate with adaptive binning and proper scoring rules, which agree in
635 direction. The max–min coverage gap is sensitive to the number and size of groups, which is precisely the phenomenon RQ3 documents; worst-group coverage is the more decision-relevant statistic. *Internal validity.* GBDT and MLP hyperparameters were fixed at common default values with no per-dataset tuning; a larger tuning budget could narrow the calibration and
640 efficiency gaps. Five seeds bound our resolution; all headline effects are nonetheless significant at $p < 10^{-3}$ under paired tests. Temperature scaling and conformal quantiles share the same calibration split, a standard but not innocuous choice. *Statistical validity.* Wilcoxon tests at $n = 15$ have a floor of $p = 6.1 \times 10^{-5}$; Friedman ranks over five models and 15 cells are
645 descriptive. We did not run distribution-shift experiments (e.g., cross-state transfer), so exchangeability holds by construction here and coverage under shift is untested. *Theoretical scope.* The noise floor of Eqs. (8)–(9) is a leading-order estimate: it assumes approximate independence of per-group coverages and uses a Gaussian-range heuristic for the expected max – min,
650 and it predicts the *order of magnitude* and *sign* of the marginal-to-Mondrian

change rather than an exact value. A fully rigorous finite-sample bound on the gap statistic—accounting for the Beta law of conformal coverage and the dependence among groups sharing a label space—is left to future work; our claim is only that the heuristic suffices to call the regime on every axis we study. *Model specificity.* The audited model is one specific public checkpoint (TabPFN v3, `tabpfn-v3-classifier-v3_default.ckpt`); conclusions about “TabPFN” are conclusions about that release and may not transfer to other versions, ensembling settings, or the regression variant. *Ethical scope.* Equalizing coverage is one narrow fairness criterion; it neither implies nor substitutes for fairness of the underlying decisions, and the race categories in ACS data are coarse administrative constructs.

8. Conclusion and future work

We showed that marginal conformal validity is not subgroup validity. Across five predictors and three census tasks, split conformal prediction met its marginal coverage guarantee while concealing an eight-point age coverage gap common to all models; group-conditional (Mondrian) calibration repaired this where strata were large but *backfired* on the small-support race axis—a concrete, statistically significant demonstration that equalized coverage in expectation can coexist with larger realized disparities. These effects were governed by the conformal procedure and the group structure rather than the model, and a simple finite-sample variance argument ($\sqrt{\alpha(1 - \alpha)/n_g}$) predicts which regime obtains. We also documented a silent failure of non-randomized APS on binary tasks. As the timely model under audit, TabPFN proved the most reliable base learner we tested: its native probabilities were

675 as calibrated as a well-regularized MLP and far better than untuned GBDTs,
 required no temperature scaling, and yielded the smallest conformal predic-
 tion sets at every coverage level. Together these results give the first matched-
 protocol picture of a tabular foundation model’s uncertainty *and* a clean
 characterization of when group-conditional conformal calibration should be
 680 trusted.

Future work proceeds on four fronts. (i) A Tier-B OpenML extension with
 a Friedman–Nemenyi ranking across heterogeneous tabular domains would
 test the generality of the efficiency ordering and, more importantly, populate
 the $(\min_g n_g, |\mathcal{G}|)$ plane with many more points to stress-test the noise-floor
 685 predictor of Section 3.5. (ii) Distribution-shift stress tests—cross-state and
 temporal transfer within ACS—would probe coverage degradation when ex-
 changeability fails, together with adaptive remedies [22, 23]. (iii) A multiclass
 study would re-open the APS question: deterministic APS is degenerate only
 because the binary label space lets the true-label cumulative mass saturate
 690 at 1, and we expect randomized or label-shared variants [13, 14] to behave
 very differently with more classes, where adaptivity to conditional coverage
 matters most. (iv) A clinical-tabular extension, where TabPFN’s small-data
 strengths and the cost asymmetry of miscoverage make equitable uncertainty
 both harder and more consequential, and where small protected strata will
 695 make the Mondrian failure mode we identified the central methodological
 obstacle.

Beyond the specific findings, the paper offers a transferable lesson for
 uncertainty quantification under group structure: an in-expectation guar-
 antee should be reported together with the finite-sample conditions under

700 which it is informative. The noise floor of Eq. (9) is one such condition—
 computable from the calibration counts alone, model-free, and validated here
 on three protected axes—and we expect analogous floors to govern other
 group-conditional procedures (class-conditional conformal prediction, multi-
 valid calibration, group-wise recalibration) wherever a guarantee is bought
 705 with per-group data. Making that purchase price explicit is, we argue, the
 right default for deploying set-valued predictors in consequential, demograph-
 ically structured settings, and it is the methodological stance we hope this
 audit advances.

Reproducibility statement

710 All code, notebooks (smoke test, experiment runner, analysis), and the
 complete per-seed result files (`calibration.csv`, 75 rows; `conformal_fairness.csv`,
 3,600 rows) are released at <https://github.com/Nadim-Mahmud-23/tabpfn-uncertainty-audit>
 under the MIT license, with the following pinned dependency versions: `tabpfn`
 8.0.7 (PyTorch 2.9.1 backend, classifier checkpoint `tabpfn-v3-classifier-v3_default.ckpt`),
 715 `xgboost` 3.2.0, `lightgbm` 4.6.0, `scikit-learn` 1.8.0, `folktables` 0.0.12,
`numpy` 2.3.5, and `scipy` 1.17.0 under Python 3.11+. Every table and statistic
 in this paper is regenerable from the committed CSVs by the analysis scripts;
 figures regenerate from the same files. Data are downloaded automatically
 from the US Census Bureau via `folktables`; no manual data placement is
 720 required. All experiments run on a single Apple Silicon MacBook with no
 GPU (CPU/MPS backend); the full experiment matrix completes in approx-
 imately one hour of wall-clock time (about 55 minutes for the 75-cell run,
 dominated by TabPFN’s CPU inference), well within the reach of commodity

hardware.

725 **Declaration of competing interest**

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

730 **Nadim Mahmud Dipu:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Data availability

The data used in this study are publicly available from the U.S. Census
735 Bureau American Community Survey (ACS) Public Use Microdata Sample (PUMS), 2018 1-Year release, accessed programmatically via the folkttables package [11]. All source code, notebooks, and the committed per-seed result files needed to reproduce every table and figure are released in the repository cited in the Reproducibility statement.

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